RectifiedTraj Revisited

An Expanded Evaluation of Map-Free Causal GPS Trajectory Denoising

Dexin Ren[1], Shang-Ling Hsu[1], Chen Chu[1], Lifeng Lin[2], Ryosuke Shibasaki[3], Noboru Koshizuka[2], Cyrus Shahabi[1]

[1]University of Southern California
[2]University of Tokyo
[3]Reitaku University



Mobile GPS Trajectory Denoising

Mobile GPS data:

- **Mobile GPS Noise sourced from**: weak reception, signal reflections, and environmental noise
- **GPS noise propagate to downstream application:** navigation, geofencing, and mobility analysis.

How can we accurately denoise GPS observations in real time on mobile devices? What key features this question require?

- **Prediction should not rely on future observations** → **Causal / Online**
  - Use only current and past observations.
- **Maps and network services may be unavailable** → **Map-Free**
  - No external infrastructure required.
- **Mobile computing resources are limited** → **Lightweight**
  - Low-latency, CPU-only inference.







Related Work

Causal

- **Global smoothers:** prediction use proceeding observations

Map-Free

- **Map matching:** requires map data, assumes road-centered trajectories

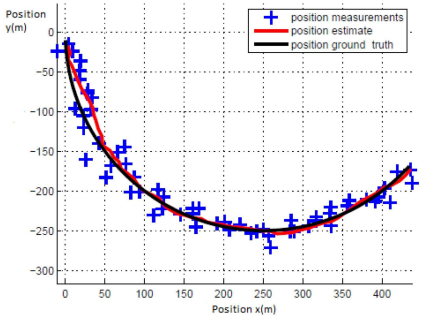
Lightweight

- **Diffusion models:** multi-step inference, expensive

What already satisfies all three?

- **Classical filters:** causal, map-free, lightweight
  - **But still leave large residual errors**

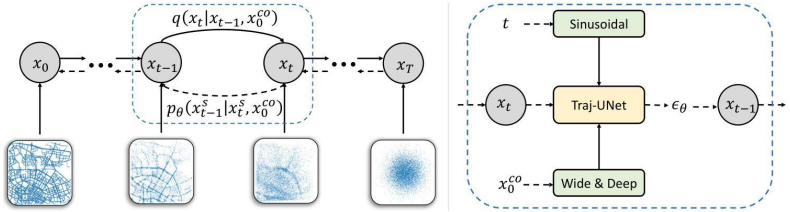
Research Gap: Accurate GPS denoising that is simultaneously **causal**, **map-free**, and **lightweight**.



Filters & smoothers



Map-matching



Diffusion model

Rectified Flow

Rectified Flow connects clean (X_0) and noisy (X_1) trajectories through a straight interpolation path:

$$X_\tau = (1-\tau)X_0 + \tau X_1$$

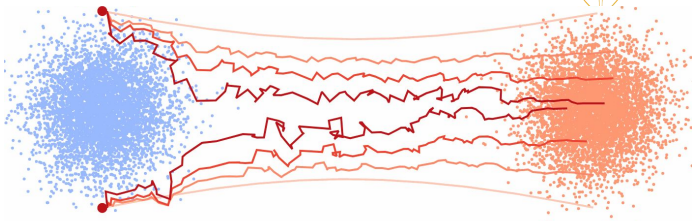
$\tau \in [0,1]$ represents the noise level.

The ideal velocity remains constant along this path. We train the model at different noise levels to predict this velocity:

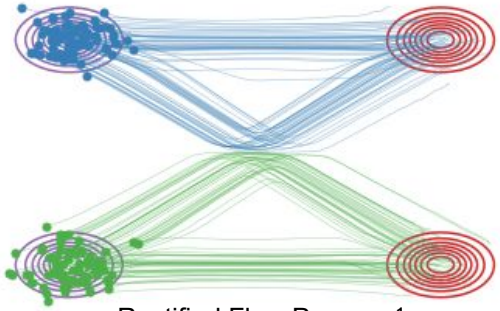
$$v_\theta(X_\tau, \tau) \approx X_1 - X_0$$

Since the path is straight, we can directly correct the noisy trajectory with one network evaluation:

$$\hat{X}_0 = X_1 - v_\theta(X_1, 1)$$



Diffusion Process [1]



Rectified Flow Process [1]

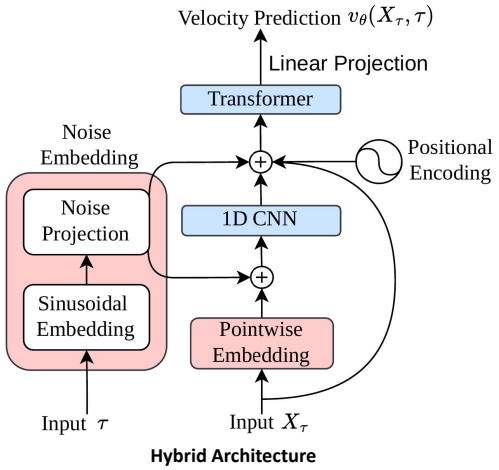


[1] Figure source: Liu et al., Flow Straight and Fast (ICLR 2023) https://rectifiedflow.github.io/

Model Architectures & Pipeline

Hybrid Architecture (Causal):

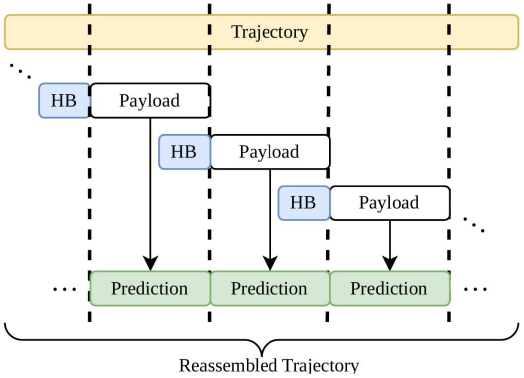
- Explore how local + long-range context affects GPS denoising quality.
- Causal CNN layer (explore aggregated local context)
- Causal Transformer layer (explore long-range context)
- Pointwise linear projection



Hybrid Architecture

Chunking System

- 256-point fixed-size chunks
- Q-point overlap (HB)
- Discard first Q predictions
  - Lack of past observations, high error region
- Concatenate retained predictions



Datasets

NUMOSIM (Train & Test):
- Synthetic data
- Gaussian noise
- Sample time:
  - 10s (train), 30s, 1min, 2min

Pattern Of Life (Test only):
- Synthetic data
- Ornstein–Uhlenbeck noise
- Sample time:
  - 5s, 10s, 30s, 1min, 2min

BlogWatcher (Test only):
- Real world data
- Real GPS noise
- Sample time:
  - Median: 10s
  - Mean: 116.44s
  - P95: 303.0s

Evaluation

Evaluation Volume:
- 200 trajectories per test
- 5,000 points per trajectory

Evaluation Metric:
- Exact error (NUMOSIM, PoL)
- Accuracy (BlogWatcher)

Cross-Domain Evaluation:
- Cross noise distribution
- Cross simulator / geography
- Cross sampling interval

Strict Streaming Evaluation:
- Predict with streaming data

Mobile-Device Evaluation:
- Evaluate on Google Pixel 4a

Baselines

Classic Filters & Smoothers:
- Kalman filter, Kalman-RTS
- Alpha-Beta, Causal Hampel,
- Savitzky–Golay, Hampel,

Map Matching:
- Valhalla Meili

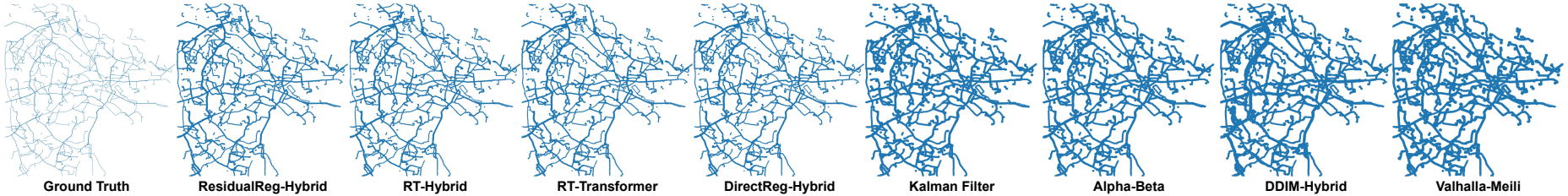
Regression:
- DirectReg-Hybrid
- ResidualReg-Hybrid
- ResidualReg-Causal MLP

Diffusion: DDIM

RectifiedTraj (RT):
- RT Transformer
- RT CNN, RT Hybrid

Performance (Synthetic Dataset)

	NUMOSIM-Kanto								Patterns of Life (PoL)									
	10s (native)		30s		1min		2min		5s (native)		10s		30s		1min		2min	
Method	Mean ↓	Online ↓	Mean ↓	Online ↓	Mean ↓	Online ↓	Mean ↓	Online ↓	Mean ↓	Online ↓	Mean ↓	Online ↓	Mean ↓	Online ↓	Mean ↓	Online ↓	Mean ↓	Online ↓
Raw trajectory	20.22	20.22	19.92	19.92	19.92	19.92	19.95	19.95	36.77	36.77	37.24	37.24	37.19	37.19	37.21	37.21	37.23	37.23
Alpha-Beta†	13.94	13.94	19.76	19.76	19.78	19.78	19.84	19.84	36.71	36.71	37.24	37.24	37.11	37.11	37.03	37.03	36.96	36.96
Causal Hampel†	18.91	18.91	22.03	22.03	25.54	25.54	33.98	33.98	36.28	36.28	40.43	40.43	39.84	39.84	41.66	41.66	54.46	54.46
Kalman Filter†	14.99	14.99	19.66	19.66	19.82	19.82	19.95	19.95	37.04	37.04	37.65	37.65	37.32	37.32	36.85	36.85	36.80	36.80
Hampel	18.84	N/A	18.59	N/A	18.66	N/A	18.82	N/A	36.54	N/A	37.46	N/A	36.24	N/A	35.44	N/A	35.05	N/A
Kalman-RTS	9.02	N/A	17.01	N/A	18.38	N/A	20.20	N/A	34.92	N/A	37.40	N/A	34.87	N/A	32.82	N/A	33.35	N/A
Savitzky–Golay	9.68	N/A	12.72	N/A	17.36	N/A	26.40	N/A	34.75	N/A	36.58	N/A	31.88	N/A	29.53	N/A	33.39	N/A
Valhalla Meili	30.34	N/A	29.40	N/A	29.07	N/A	29.18	N/A	44.81	N/A	42.95	N/A	41.72	N/A	41.10	N/A	40.46	N/A
Causal MLP†	5.42	5.42	7.19	7.19	9.68	9.68	12.35	12.35	24.19	24.19	25.06	25.06	25.48	25.48	27.30	27.30	31.61	31.61
DirectReg-Hybrid†	6.61	19.63	102.84	256.69	283.94	563.32	393.15	784.76	**17.74**	12.53	18.98	20.25	20.15	25.30	22.46	33.82	30.14	60.03
ResidualReg-Hybrid†	**3.01**	**2.07**	3.44	2.86	4.22	4.10	5.02	5.45	27.14	24.40	22.67	19.84	12.83	10.97	**9.87**	**9.08**	**10.77**	**12.13**
DDIM-Hybrid†	18.15	N/A	839.56	N/A	2566.16	N/A	3566.32	N/A	22.29	N/A	26.33	N/A	28.89	N/A	32.07	N/A	41.16	N/A
RT-CNN†	4.49	4.50	5.22	5.87	6.33	7.53	7.35	9.13	26.52	25.96	24.37	24.58	18.06	19.52	15.49	18.09	17.12	22.27
RT-Trans.†	3.26	2.55	3.96	3.92	5.05	5.68	6.08	7.45	19.21	**11.79**	**16.24**	**12.82**	**12.22**	11.93	11.64	13.31	14.75	20.39
RT-Hybrid†	3.04	2.12	**3.43**	**2.85**	**4.17**	**4.07**	**4.84**	**5.23**	24.25	21.19	20.20	17.30	12.44	**10.66**	10.16	9.72	11.25	13.18



Performance (Real World - BlogWatcher)

Method	All		≤30m		≤15m		≤10m		≤5m	
	Acc. ↑	ΔAcc. ↑	Acc. ↑	ΔAcc. ↑	Acc. ↑	ΔAcc. ↑	Acc. ↑	ΔAcc. ↑	Acc. ↑	ΔAcc. ↑
Raw data	44.63%	0.00	43.45%	0.00	42.82%	0.00	38.52%	0.00	36.42%	0.00
Alpha-Beta†	36.33%	-8.30	32.11%	-11.34	29.62%	-13.20	19.28%	-19.24	15.32%	-21.10
Causal Hampel†	37.90%	-6.73	37.20%	-6.25	36.36%	-6.46	32.47%	-6.05	30.17%	-6.25
Kalman Filter†	43.52%	-1.11	42.20%	-1.25	41.49%	-1.33	36.95%	-1.57	34.84%	-1.58
Hampel	45.46%	+0.83	44.07%	+0.62	43.37%	+0.55	38.77%	+0.25	36.44%	+0.02
Kalman-RTS	35.72%	-8.91	33.55%	-9.90	32.18%	-10.64	26.37%	-12.15	23.57%	-12.85
Savitzky–Golay	35.43%	-9.20	31.08%	-12.37	27.69%	-15.13	16.06%	-22.46	10.89%	-25.53
Valhalla Meili	50.80%	+6.17	48.90%	+5.45	48.16%	+5.34	44.47%	+5.95	44.24%	+7.82
Causal MLP†	45.18%	+0.55	42.83%	-0.62	40.87%	-1.95	30.57%	-7.95	27.77%	-8.65
DirectReg-Hybrid†	21.81%	-22.82	19.00%	-24.45	17.02%	-25.80	8.96%	-29.56	5.78%	-30.64
DDIM-Hybrid†	18.29%	-26.34	15.54%	-27.91	13.78%	-29.04	6.22%	-32.30	4.29%	-32.13
RT-CNN†	55.17%	+10.54	53.61%	+10.16	51.28%	+8.46	45.19%	+6.67	42.10%	+5.68
RT-Trans.†	55.51%	+10.88	54.22%	+10.77	52.13%	+9.31	**47.12%**	**+8.60**	**44.74%**	**+8.32**
RT-Hybrid†	**57.05%**	**+12.42**	**54.64%**	**+11.19**	**52.15%**	**+9.33**	45.40%	+6.88	42.30%	+5.88

*ResidualReg-Hybrid evaluation on BlogWatcher could not be completed within the study period.

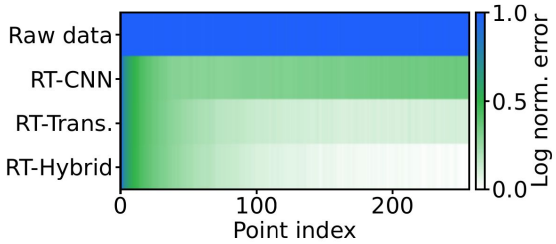
Error Distribution & Reduction

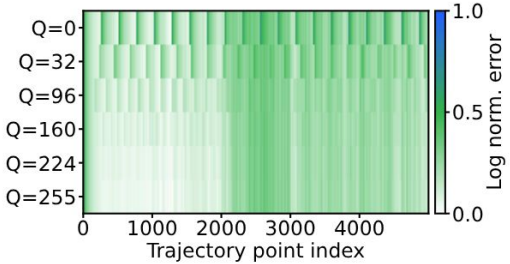
Error Distribution:

- Models using long-range context (RT-Transformer and RT-Hybrid) show higher prediction errors at the beginning of each chunk.
- Errors gradually decrease as more historical context becomes available.

Error Reduction (Head-Buckle):

- Discard the first Q predictions from each chunk and retain the remaining predictions.
- Larger Q generally reduces prediction error but also decreases throughput, introducing an accuracy–throughput trade-off.





Q points	Mean ↓	Tail ↓	p95 ↓	retained/s ↑
0	3.04	2.10	8.83	**37,832**
16	2.68	2.11	6.48	35,599
32	2.55	2.16	5.77	33,232
48	2.44	2.11	5.36	31,078
80	2.34	**2.06**	4.94	26,699
128	2.24	2.06	4.66	19,361
192	2.16	2.06	**4.56**	9,769
240	2.13	2.09	4.60	3,244
255	**2.12**	2.12	4.63	249

Mobile-Device Inference

Mobile Platform:

- Google Pixel 4a (Snapdragon 730G)
- CPU-only inference, without GPU or external services

Inference Latency (256-point chunk):

- RT-CNN: 11.02 ms
- RT-Transformer: 53.82 ms
- RT-Hybrid: 124.24 ms

Deployment Feasibility:

- All three RectifiedTraj architectures complete inference well below the shortest evaluated GPS sampling interval (5s).
- Model sizes: 4.20–5.12 MB.
- Diffusion model requires at least 50 inference steps, making it substantially slower than one-step RectifiedTraj.

Table 7: Model size and forward-pass latency on a Google Pixel 4a. Size is reported in megabytes (MB) and latency in milliseconds. The p90 value is the 90th-percentile latency. Throughput is input points processed per second; DDIM reports one denoising step.

Model	Size	Mean ↓	p90 ↓	input pts./s ↑
RT-CNN	4.20	11.020	12.435	23,230
RT-Trans.	4.26	53.818	59.443	4,757
RT-Hybrid	5.12	124.235	127.130	2,061
DDIM (one step)	5.12	120.958	128.966	2,116

Conclusion

- Strong denoising performance across synthetic and real-world datasets.
- Correction-field prediction is the primary source of stability; head-buckle reassembly further reduces chunk-start errors.
- Lightweight and fast on mobile devices, with one-step inference much faster than diffusion model.

- Evaluate denser real-world GPS trajectories with exact ground truth. Test across more mobile devices and sensing environments.